

SBC Group Project

Forecasting GDP per Capita of FRANCE

Submitted By

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1. Introduction

France is one of the largest economies in the world and one of the founding members of the European Union. Its economy is largely service based. Over the study period of 1976 to 2024, we found that France experienced generally a stable upward trend improving over time in GDP per capita over the whole period, reflecting a smooth and steady long run economic growth. However, there are two notable points that are visible in the data – which include the 2008 global financial crisis and the COVID-19 pandemic in 2020 both of which caused temporary but significant fall in the GDP trend.

This paper applies time series forecasting techniques to the annual GDP per capita data of 49 yearly observation of France which is obtained from the World Bank database, covering the period 1976 to 2024 in constant 2015 US dollars. In this project we used six forecasting methods and compared — the Mean, Naive, Drift, Exponential Smoothing (ETS), STL decomposition with Drift, and ARIMA — evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). The results show that ARIMA (0,1,1) with drift performed well among all other models and is used to produce a 10-years of forecasting of France's GDP per capita from 2025 to 2034.

2. Data Description and Visualization

2.1 Data Sources

The data used in this study is annual GDP per capita for France, obtained from the World Bank database. The series is measured in constant 2015 US dollars, which adjusts for inflation. The dataset covers the period from 1976 to 2024 (total 49 observations).

2.2 Descriptive Statistics

The mean GDP per capita over the period is approximately \$31,328.37 with a minimum of \$20,700.62 in 1976 and a maximum of \$39,683.40 in 2024, showing the substantial economic growth France experienced over nearly five decades. The standard deviation indicates moderate variability around the mean, driven largely by the long run upward trend rather than random fluctuations.

2.3 Time Series Plot

Fig.1 plots France's GDP per capita over the full sample period from 1976 to 2024. The graph shows an upward trend throughout the period. Two notable disruptions are visible - a slowdown around 2008 to 2009 corresponding to the global financial crisis, and a sharp contraction in 2020 caused by the COVID-19 pandemic. In both cases the economy recovered relatively quickly and returned to its long run growth trajectory. Outside of these shocks the series grows at a broadly stable and consistent rate, suggesting that a trend-based forecasting model would be appropriate for this data.

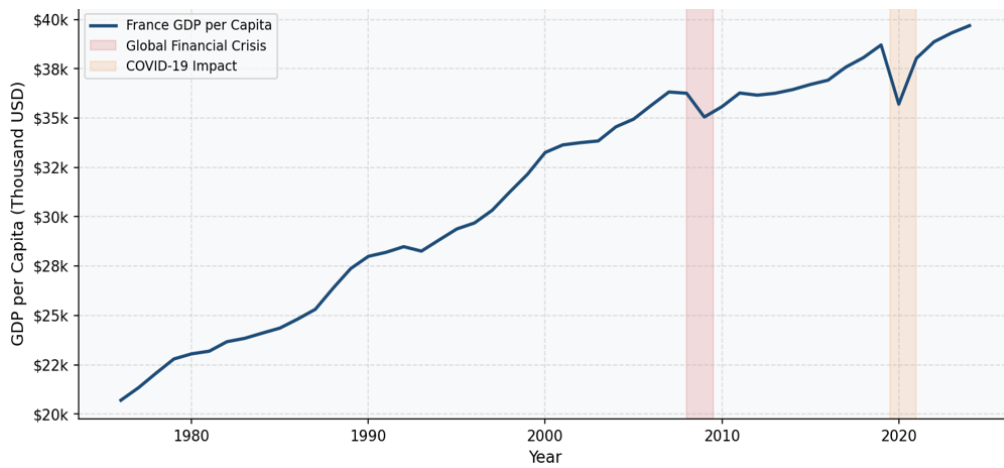


Fig.1

The Main Features of this Time Series

Trend: Looking at the trend, France's GDP per capita went up over the whole period. It started at around \$20700 in 1976 and reached about \$39683 in 2024. That means it nearly doubled in 49 years, which tells us the French economy improved during that time.

Volatility:

The fluctuations in GDP per capita are relatively moderate, indicating moderate but stable volatility, which is typical of a large, developed economy like France. GDP per capita increased smoothly over most of the sample period, without extreme short-run fluctuations or sharp downturns.

Structural Changes: Two notable disruptions are visible in the data:

- **Global Financial Crisis (2008–2009):** GDP per capita fell from \$36,322 in 2007 to \$35,053 in 2009, a decline of approximately 3.5%, reflecting France's exposure to the global banking and credit crisis.
- **COVID-19 Pandemic (2020):** This was the sharpest drop in the entire dataset, with GDP per capita falling from \$38,705 in 2019 to \$35,709 in 2020, a decline of approximately 7.7%. A rapid recovery followed by 2022 confirming the resilience of the French economy.

2.4 Train/Test Split

The dataset is divided into a training set and a test set. The training set covers the period 1976 to 2014(39 observations). The test set covers the period 2015 to 2024(10 observations). This split ensures a sufficiently long training period to capture France's historical growth patterns while providing a meaningful test horizon for model comparison. The forecast horizon is therefore $h = 10$ years.

3. Methodology

This section describes the six forecasting methods applied to France's GDP per capita. All models are estimated on the training set (1976–2014) and evaluated on the test set (2015–2024).

3.1 Mean Method

The Mean method forecasts all future values as the historical average of the training data. It assumes that the series fluctuates around a constant mean with no trend. The forecast is given by:

$$\hat{y}_{T+h|T} = \bar{y}$$

where $\bar{y}$ is the mean of all training observations. Given that France's GDP per capita exhibits a strong upward trend, this method is expected to perform poorly as it completely ignores the trend component.

3.2 Naive Method

The Naive method forecasts all future values as the last observed value in the training set. It assumes the series follows a random walk with no drift. The forecast is given by:

$$\hat{y}_{T+h|T} = y_T$$

where y_T is the last observed value. This method is appropriate for series with no trend but is expected to underperform for France's GDP per capita since it ignores the consistent upward growth pattern.

3.3 Drift Method

The Drift method extends the last observed value by adding the average change observed over the training period. It captures the long run linear trend of the series. The forecast is given by:

$$\hat{y}_{T+h|T} = y_T + h\left(\frac{y_T - y_1}{T - 1}\right)$$

where y_1 is the first observed value, y_T is the last, T is the length of the training set and h is the forecast horizon.

3.4 Exponential Smoothing (ETS)

In exponential smoothing, the weights to past observations decrease exponentially, whereas the more recent observations receive higher weights. For this analysis, Holt's linear trend method is used with an additive trend component with no seasonality. By minimising the SSE (sum of squared errors), the model automatically estimates the smoothing parameters α (level) and β (trend) from the training data set.

3.5 STL Decomposition with Drift

Seasonal and Trend decomposition using Loess (STL) decomposes the time series into trend, seasonal and residual components. Since the data is annual, a minimum period of 2 is specified, which means STL cannot detect any meaningful seasonal cycle. As expected, the seasonal component extracted from France's GDP per capita is negligibly small and oscillates around zero, confirming that annual GDP data has no meaningful seasonality. The decomposition is shown in Figure 2.

After decomposition, the seasonal component is removed from the training series, and a Drift forecast is applied to the de-seasonalised series. The projected seasonal component is then added again to obtain the final forecast. While STL adds limited value for annual data as compared to a pure Drift model.

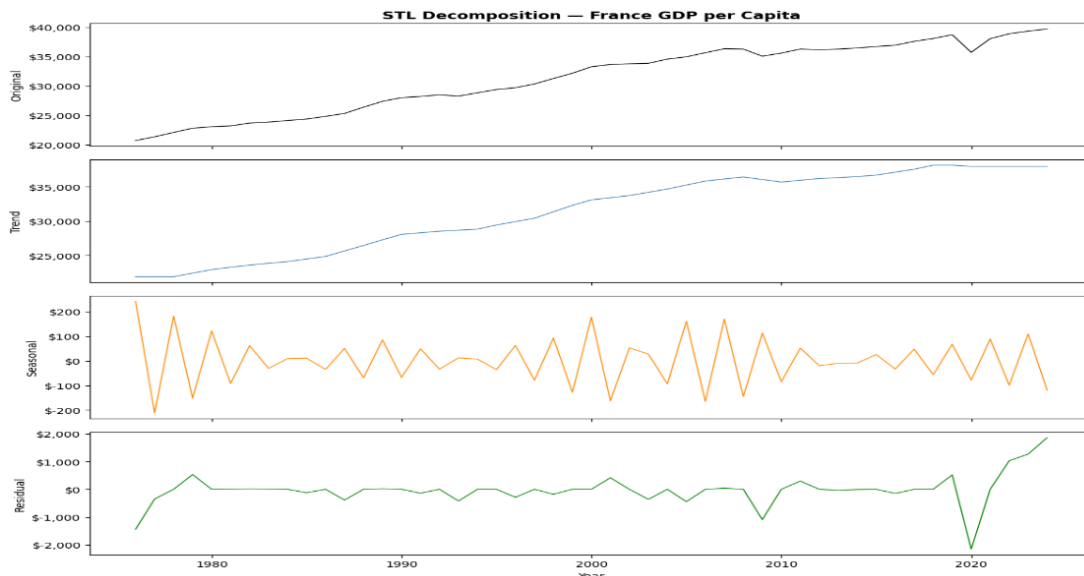


Fig.2

3.6 ARIMA

3.6.1 Transformation Check

Before fitting an ARIMA model, the Box-Cox procedure is used for variance-stabilising transformation. The estimated λ value is 1.16, (close to 1), indicating that the variance is stable over time and no transformation is needed. Therefore, the raw series is used directly in the ARIMA modelling process.

3.6.2 Stationarity Tests

ARIMA requires the time series to be stationary. Stationarity was assessed using two complementary tests - the Augmented Dickey-Fuller (ADF) test and the KPSS test. The results are presented in Table 1.

SERIES	ADF P-VALUE	KPSS P-VALUE	CONCLUSION
RAW GDP (D=0)	>0.05	<0.05	Non-Stationary
FIRST DIFF (D=1)	<0.05	>0.05	Stationary

Table 1: Stationarity Test Results

The raw GDP series was found to be non-stationary. After applying first differencing ($d=1$), both tests confirmed stationarity. Therefore $d=1$ was selected as the order of integration. Figure 3 plots the raw and first-differenced series.

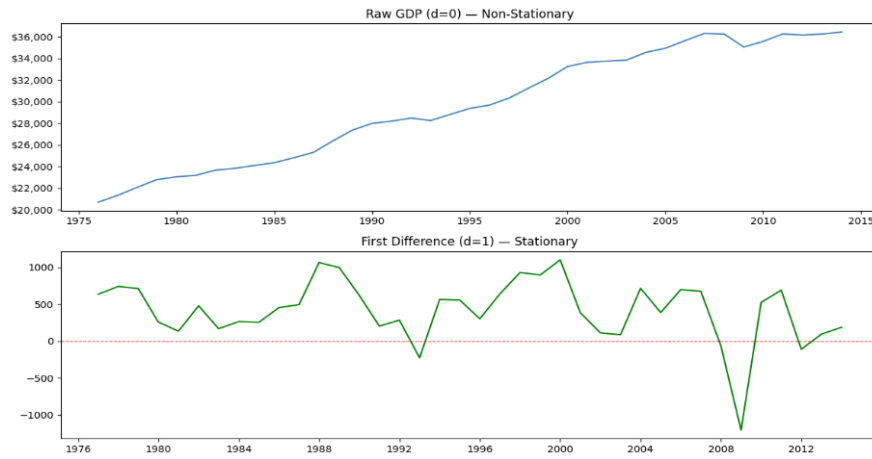


Fig.3

3.6.3 ACF and PACF Analysis

Both the ACF and PACF show marginal spikes at lag 1 that fall within or on the boundary of the 95% confidence band, with no significant spikes at higher lags. Based on this analysis, four candidate models were identified — ARIMA(0,1,0), ARIMA(0,1,1), ARIMA(1,1,0) and ARIMA(1,1,1).

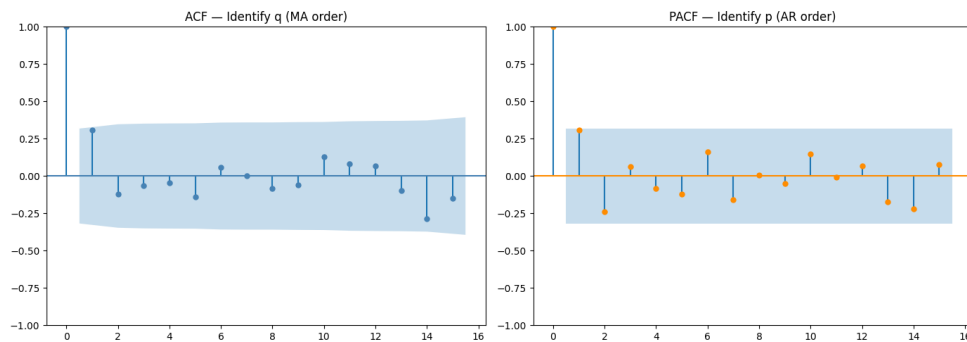


Fig.4

3.6.4 Model Selection

The Akaike Information Criterion (AICc), which penalises model complexity and is used to compare four candidate models. As results presented in Table 2. ARIMA (0,1,1) with drift achieves the lowest AICc value and it is also confirmed as the best order by Auto ARIMA. The consistency between the manual comparison and Auto ARIMA strongly validates this selection. The AICc differences between the top models are small, indicating that all top candidates are similarly competitive, but only ARIMA(0,1,1) with drift is preferred.

MODEL ORDER	AICC	AIC	BIC
ARIMA(0, 1, 1) + DRIFT	570.091	569.385	574.298
ARIMA(1, 1, 0) + DRIFT	570.293	569.587	574.500
ARIMA(0, 1, 0) + DRIFT	570.833	570.490	573.765
ARIMA(1, 1, 1) + DRIFT	572.522	571.310	577.860

Table 2

3.5.5 Model Interpretation

The differencing order $d=1$ confirms the fact that France's GDP per capita grows at a constant rate rather than accelerating. The absence of autoregressive terms ($p=0$) indicates that GDP levels in the past, beyond one period, do not add meaningful predictive power once the trend is accounted for. The moving average term ($q=1$) captures the short run effect of unexpected economic shocks from one year to the next. The long run average annual growth is captured by drift term.

3.5.6 Residual Diagnostics

The residual diagnostics are conducted in order to validate the ARIMA model. The residuals fluctuate randomly around a zero mean with constant variance over the time plot, indicating a stable model fit. There are no significant spikes as presented by the ACP plot. The Ljung-Box test ($p = 0.823$, p -value greater than 0.05) confirms that the residuals are white noise, meaning all temporal patterns have been captured. The Histogram and Q-Q plot shows that the residuals are approximately normally distributed.

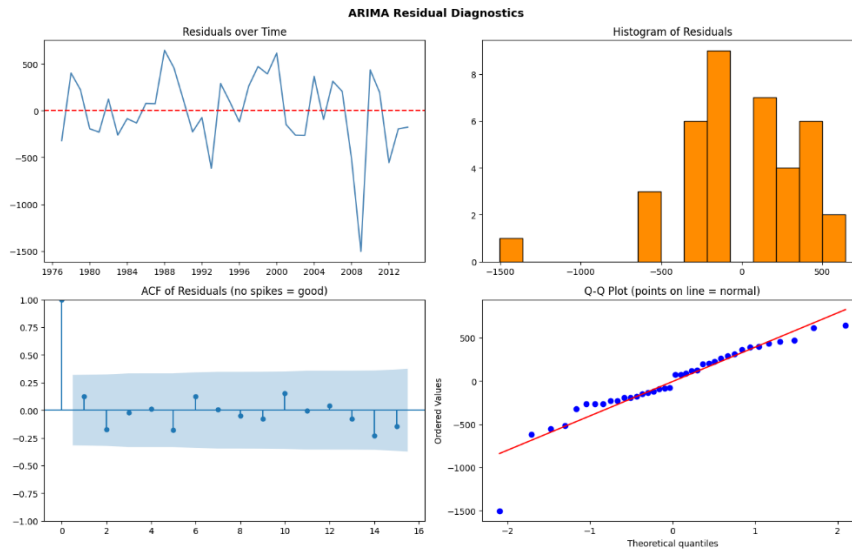


Fig.5

4. Results and Model Comparison

4.1 Forecast Comparison

All six models are fitted on the training set (1976–2014) and used to generate forecasts for the test period (2015–2024). The forecast comparison is shown in Fig. 6, where the left panel shows the full historical series along with all the model forecasts and the right panel shows the test period in zoomed form for better comparison.

The Mean method performs the worst as shown by the right panel. The forecasting line is flat at approximately \$29,000 which is far below the actual test values. In the same manner the naive method underperforms by carrying forward the last training value of approximately \$36,000 with no growth. The Drift, ETS and STL with Drift methods all capture the upward trend and produce similar forecasts that follow the direction of the actual series. ARIMA(0,1,1) with drift tracks the actual values most closely throughout the test period, while partially capturing the COVID slowdown in 2020.

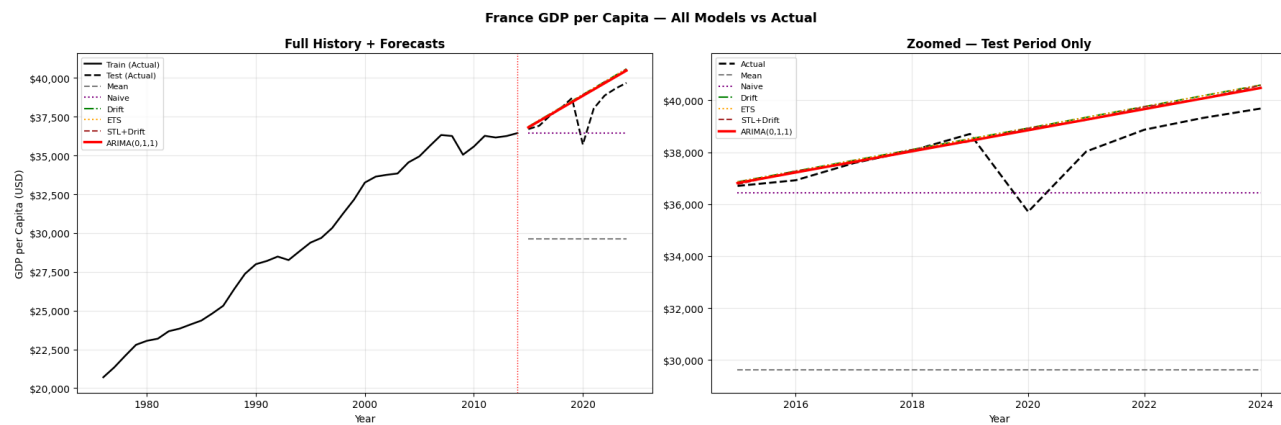


Fig.6

4.2 Forecast Accuracy

MODEL	MAE	RMSE	MAPE (%)
ARIMA(0,1,1)	745.30	1,155.69	2.00%
STL+DRIFT	768.41	1,188.76	2.06%
ETS	802.60	1,209.76	2.15%
DRIFT	802.71	1,209.87	2.15%
NAIVE	1,660.88	1,923.62	4.30%
MEAN	8,329.83	8,413.96	21.87%

Table 3: Forecast accuracy metrics

ARIMA (0,1,1) with drift has the lowest error across all metrics (MAPE: 2.00%), showing that it is the best-performing model. The Mean method has the highest error (MAPE: 21.87%) because it fails to account for the upward trend. Drift, STL+DRIFT, and ETS models have almost same error. This is because GDP per capita data doesn't show seasonal pattern, making STL decomposition unnecessary, and leading the ETS model to simplify into a basic linear trend. ARIMA(0,1,1) with drift outperformed these models because it combines the long-run growth with a moving average term that adjust for short term shocks. This behaviour of ARIMA allows for a more accurate fit.

5. Future (10 year) Forecast

After identifying the ARIMA (0,1,1) with drift as the best model, we re-estimated it using the full dataset (1976-2024) so that model can capture the historical pattern in the data and then forecast. Figure 7 presents the 10-year forecast of France's GDP per capita from 2025 to 2034, along with 95% prediction intervals.

YEAR	FORECAST	LOWER 95%	UPPER 95%	YEAR	FORECAST	LOWER 95%	UPPER 95%
2025	40,080.10	38,737.14	41,423.06	2030	42,057.47	38,887.56	45,227.38
2026	40,475.57	38,617.48	42,333.66	2031	42,452.95	39,032.81	45,873.08
2027	40,871.05	38,612.41	43,129.69	2032	42,848.42	39,195.17	46,501.68
2028	41,266.52	38,668.37	43,864.68	2033	43,243.90	39,371.53	47,116.26
2029	41,662.00	38,763.83	44,560.17	2034	43,639.37	39,559.64	47,719.10

Table 4

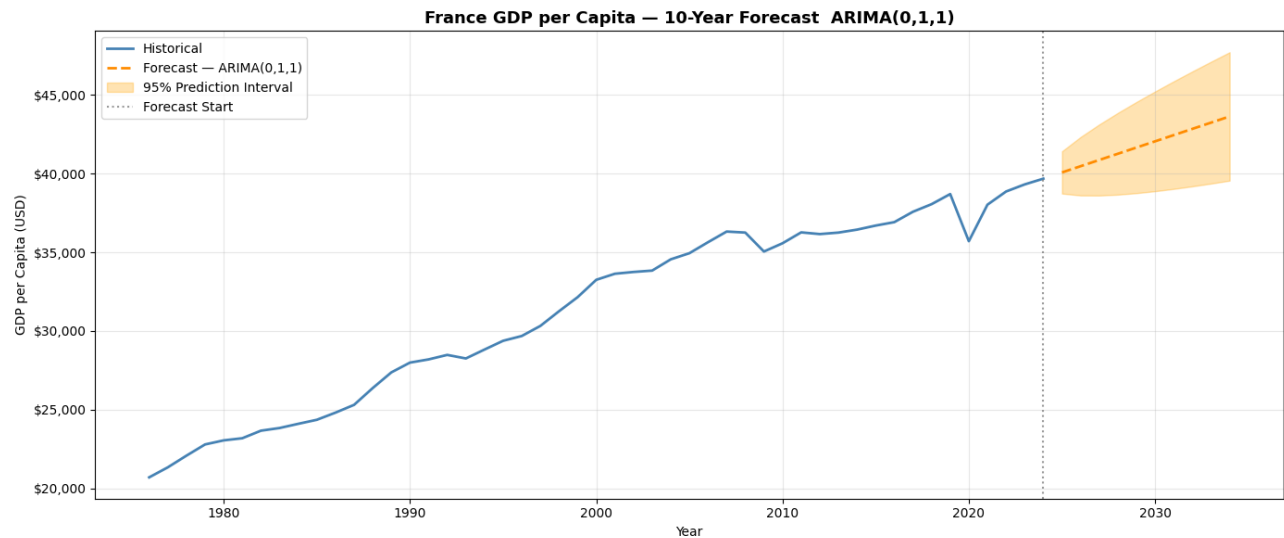


Fig.7

Forecast graph shows that there is a steady and consistent increase in France's GDP per capita from approximately \$40,080 in 2025 to \$43,639 by 2034, reflecting an average annual growth of approximately \$395. This generally aligns with France's typical growth pattern seen during the training period. As we look further into the future, the prediction intervals become wider, which is quite natural since uncertainty increases over time. By 2034, the forecast range stretches from about \$39,560 to \$47,719, highlighting the inherent uncertainty involved in long-term economic projections.